



ABHEY TIWARI

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PROFESSIONAL SUMMARY

AI/ML engineer focused on building production-ready LLM systems, RAG pipelines, local inference with Ollama, agentic architectures using LangGraph, and real-time backends with FastAPI and WebSockets. Experienced deploying and iterating on end-to-end AI systems, with published research in democratising AI for low-resource environments.

TECHNICAL SKILLS

AI / ML: LLMs, RAG pipelines, vector search, embeddings, NLP, CNNs, agentic systems

Frameworks: FastAPI, Flask, LangChain, LangGraph, LlamaIndex, Ollama, Streamlit

ML Tools: Scikit-learn, TensorFlow, PyTorch, ChromaDB

Languages: Python, C++

Infrastructure: Docker, Git, Linux, REST APIs, WebSockets, WebRTC

EXPERIENCE

Tech, Research & Innovation Intern: Draupadi Dream Trust Jul 2025 – Aug 2025

- Conducted multi-source research on Yamuna River (religious, cultural, environmental dimensions) and analysed Delhi Jal Board datasets.
- Built a FastAPI + Gemini PDF Q&A tool (Quizard) to accelerate document retrieval workflows during the internship.
- Produced structured stakeholder-ready reports synthesising quantitative and qualitative findings.

Frontend Developer (Remote): BeyondRiffs Sep 2024 – Dec 2024

- Rebuilt major React components prior to product launch, rapidly learning React to stabilise production.
- Coordinated with design and backend teams to ship a reliable release under tight timelines.

Website Developer: Mangalam Valley Resort Jul 2023 – Nov 2023

- Delivered a booking-enabled website with integrated payments and SSL.
- Managed hosting, domain configuration, and end-to-end deployment workflows.

PROJECTS

Maitri AI: *Real-time emotionally aware local companion*

Stack: DeepFace, Ollama, FastAPI, WebSockets, Python

- Privacy-first mental-health assistant with 100% local inference — zero cloud data transmission.
- Integrated real-time facial emotion recognition via DeepFace with context-aware dialogue generation using local LLMs through Ollama.
- Persistent emotional context maintained across conversation turns via WebSocket session state.

INDICA v1.0: *Modular voice-controlled personal cognitive assistant*

Stack: Python, Gemini API, speech_recognition, pyttsx3, asyncio, Playwright

- Built a full voice agent pipeline: STT → intent recognition → task dispatcher → skills → TTS.
- Architected a dual-memory system: short-term session logs injected as LLM context and a semantic long-term memory layer using embeddings.
- Modular dispatcher routes intents to skills (app control, web search, email, weather, code generation) without hallucinated actions.

Quizard: *AI-powered PDF Q&A tool for fast document retrieval*

Stack: FastAPI, Streamlit, Google Gemini, PyMuPDF, Python

- Built to solve a real workflow bottleneck during internship, eliminated manual search through 90+ page documents.
- Extracts and processes PDF text via PyMuPDF, feeds structured context to Gemini for accurate Q&A responses.
- Deployed as a full-stack tool with FastAPI backend and Streamlit frontend; results returned in under 3 seconds.

OutreachPro: AI-assisted batch email outreach tool (open-source)

Stack: Chrome Extension (MV3), Gmail API, Google Sheets API, Gemini API, OAuth 2.0, JavaScript

- Co-built an open-source alternative to paid cold-email platforms, lets teams send personalised batch emails via Gmail without third-party subscriptions.
- Owned and built the AI layer: Gemini-powered email template refinement that improves subject lines, body copy, and CTAs while preserving template variables.
- Integrated Gmail API rate-limiting (500/day cap, 3s send intervals) and live Google Sheets status sync.

Cancer Detection AI: *Lightweight multimodal diagnostic assistant*

Stack: TensorFlow, PyTorch, XGBoost, Python

- Hybrid CNN + XGBoost approach for MRI and skin-lesion analysis, optimised for standard laptop hardware.
- Produces interpretable predictions to support clinical decision-making in low-resource settings.

EDUCATION

BSc. Mathematical Sciences (3rd Year): Maharaja Agrasen College, University of Delhi 2023 – *Present*

Class XII (Science, PCMB): Kendriya Vidyalaya, Noida Sector 24 2023

Class X : Kendriya Vidyalaya, Noida Sector 24 2021

AWARDS & RECOGNITION

Best Paper Award — 9th National Student Academic Conference (NSAC) April 2026

Maharaja Agrasen College: *"Democratizing Intelligence: Neural Pruning, Local AI Deployment, and Open-Source Tools as Pathways to Equitable Access in the Global South"*